

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM			
SYSTEM:	Abnormal Operating Procedures			
TASK:	TCAF PZR Pressure Malfunction (Isolate leaking PZR PORV)			
TASK NUMBER:	N1140240401			
JPM NUMBER:	13-01 NRC Sim c			
ALTERNATE PATH:	☐	K/A NUMBER:	APE 027 AA1.01	
APPLICABILITY:		IMPORTANCE FACTOR:	4.0 3.9	
EO ☐	RO ☒	STA ☐	SRO ☒	
EVALUATION SETTING/METHOD:	Simulator - Perform			
REFERENCES:	S2.OP-AB.RC-0001, Rev. 11 Reactor Coolant System Leak (rev checked 10-7-14) S2.OP-AB.PZR-0001, Rev. 18 Pressurizer Pressure Malfunction (rev checked 10-7-14)			
TOOLS AND EQUIPMENT:				
VALIDATED JPM COMPLETION TIME:	<u>4 min</u>			
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>NA</u>			
Developed By:	G Gauding Instructor	Date:	9-6-14	
Validated By:	D Tait SME or Instructor	Date:	10-15-14	
Approved By:	 Training Department	Date:	10-23-14	
Approved By:	 Operations Department	Date:	10-27-14	
ACTUAL JPM COMPLETION TIME:				
ACTUAL TIME CRITICAL COMPLETION TIME:				
PERFORMED BY:				
GRADE:	☐ SAT	☐ UNSAT		
REASON, IF UNSATISFACTORY:				
EVALUATOR'S SIGNATURE:			DATE:	

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DATE: _____

SYSTEM:

TASK: N1140240401

TASK NUMBER:

INITIAL CONDITIONS: The unit is operating in a steady state condition at 100% power with all systems in automatic. 21 Charging pump is in service.

SIMULATOR SETUP: IC-253

The simulator setup is:

RT-1

MALF: PR0018B 2PR2 Develops Leak

Severity: 5000

- **Ensure a centrifugal charging pump is in service.**

INITIATING CUE:

You are the Reactor Operator.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Determine that 2PR2 PZR PORV is leaking and close 2PR7 Block Valve.

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	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
			Simulator Operator: Insert RT-1		
			Operator either: Identifies PORV tailpipe temperature has risen from normal to ~227°F OR Identifies rising charging flow THEN enters S2.OP-AB.RC-0001, Reactor Coolant system Leak <u>or</u> S2.OP-AB.PZR-0001, Pressurizer Pressure Control Malfunction		
	AB.RC 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary. Note: CAS Item 3.0 directs entry into S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction. When PORV leakage is identified, operator initiates S2.OP-AB.PZR-0001.		

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	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.2	Is RCS temperature <350°F? (Mode 4, 5, or 6 LOCA)	Answers NO and goes to Step 3.8 (Mode 1, 2, or 3 LOCA)		
	3.8	Is the Unit in Mode 3 after the accumulators are isolated?	Answers NO and goes to Step 3.12 (Mode 1, 2 or 3 RCS LEAK)		
	3.12	Is PZR level being maintained stable or rising?	Answers YES and goes to Step 3.18 Note: If operator is unsure if PZR level is lowering, then they may adjust charging flow to stabilize PZL level.		
	3.18	PLACE CVC Makeup Control System in service as necessary to control VCT level.	Operator determines Makeup Control System is in automatic.		
	3.19	INITIATE S2.OP-ST.RC-0008(Q), Reactor Coolant System Water Inventory Balance.	Cue: A second operator will initiate the RCS Water Inventory Balance.		
	3.20	IF recommended by Rad Pro OR desired to reduce containment radiation levels THEN PLACE 2 CFCUs in slow speed AND 2 CFCUs in fast speed.	Determines no recommendation by Rad Pro has been made, and containment pressure has remained stable.		
	3.21	PERFORM the following to identify location of leak:			

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	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.21.1	INITIATE actions to locate and isolate the leak IAW Attachment 2, Guidelines for Locating An RCS Leak.	Initiates actions to locate and isolate the leak IAW Attachment 2, Guidelines for Locating An RCS Leak.		
	Att. 2 1.0	MONITOR indicators on the left side of Table 1 for high or rising readings.	Monitors indicators on the left side of Table 1 for high or rising readings, and determines PORV tailpipe temperature is elevated.		
	CAS 3.0	IF AT ANY TIME , the leak is determined to be from a PORV, OR a Pressurizer Code Safety Valve, THEN INITIATE S2.OP-AB.PZR-0001(Q), Pressurizer Pressure Malfunction.	Determines leak is from a PZR PORV, and initiates S2.OP-AB.PZR-0001(Q), Pressurizer Pressure Malfunction.		
	AB.PZR 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	3.2	Is POPS in service?	Determines POPS is not in service.		
	3.3	Is the controlling Pressurizer Pressure Control Channel (I or III) failed?	Determines the controlling Pressurizer Pressure Control Channel (I or III) has not failed, and goes to Step 3.11.		
	3.11	Is the Master Pressure Controller failed? (Refer to Attachment 2 for guidance)	Determines the Master Pressure Controller has not failed, and goes to Step 3.17		
	3.17	Is a Spray Valve(s) failed? (Refer to Attachment 2 for guidance)	Determines a Spray Valve has not failed and goes to Step 3.39.		

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	3.39	Is a PORV(s) failed? (Refer to Attachment 2 for guidance)	Determines a PORV has not failed based on closed indication and goes to Step 3.46.		
	3.46	Are PORV tailpipe temperatures elevated?	Determines PORV tailpipe temperatures are elevated.		
*	3.47	CLOSE 2PR6 AND 2PR7, Pressurizer Relief Stop Valves.	Closes 2PR6 AND 2PR7, Pressurizer Relief Stop Valves.		
	3.48	IF PORV tailpipe temperature does NOT lower sufficiently to allow a temperature rise to be seen when the leaking PORV is unisolated, THEN:...	Determines PORV tailpipe temperatures are lowering sufficiently.		
*	3.49	When pressure stabilizes, OPEN 2PR6, Pressurizer Relief Stop Valve.	Determines pressure is stable, and opens 2PR6, Pressurizer Relief Stop Valve.		
	3.50	Is tailpipe temperature rising?	Determines tailpipe temp is not rising and goes to Step 3.53.		
*	3.53	OPEN 2PR7.	Opens 2PR7.		
	3.54	Is tailpipe temperature rising?	Determines tailpipe temperature rising.		
*	3.55	CLOSE 2PR7.	Closes 2PR7.		

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	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.56	NOTIFY the SM/CRS to refer to Technical Specification 3.4.5 and the Event Classification Guide for Primary Leakage.	Terminate JPM.		

TERMINATING CUE:

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TQ-AA-106-0303

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- Q 1. Task description and number, JPM description and number are identified.
- Q 2. Knowledge and Abilities (K/A) references are included.
- Q 3. Performance location specified. (in-plant, control room, or simulator)
- Q 4. Initial setup conditions are identified.
- Q 5. Initiating and terminating Cues are properly identified.
- Q 6. Task standards identified and verified by SME review.
- Q 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- Q 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 19 Date 10/15/14
" "
- Q 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: 

Date: 10/15/14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

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INITIAL CONDITIONS:

The unit is operating in a steady state condition at 100% power with all systems in automatic.

21 Charging pump is in service.

INITIATING CUE:

You are the Reactor Operator.